University of Stuttgart
Institute of electrical and optical communications

Advanced optical fiber communications Wintersemester 2025/26 Exercise 6

M.Sc. Julian Schneck



Problem 10

The silicon dioxide used in an optical fiber has a material dispersion coefficient of 17 ps/(nm km). A step index fiber has a core radius of 4 µm, a numerical aperture of 0.14, and the refractive index of the cladding is 1.44. The used wavelength is 1.55 µm.

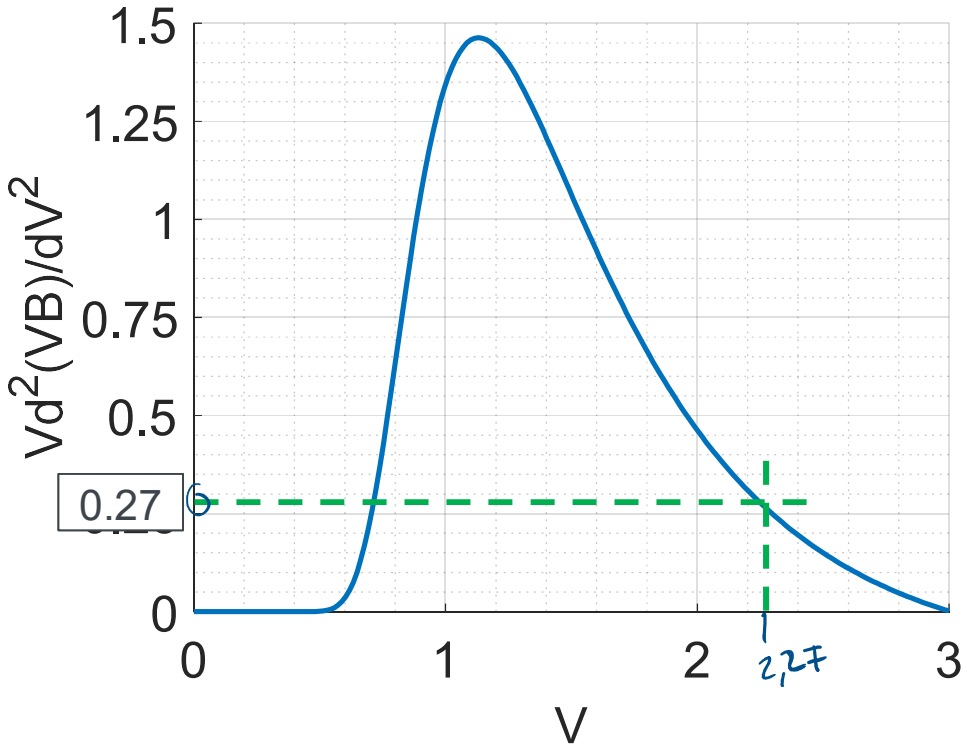
a) Calculate the total dispersion of the optical fiber.

b) Now, the total dispersion shall be minimized. Find the core radius, where the waveguide dispersion compensates the material dispersion.

Problem 10

University of Stuttgart
Germany

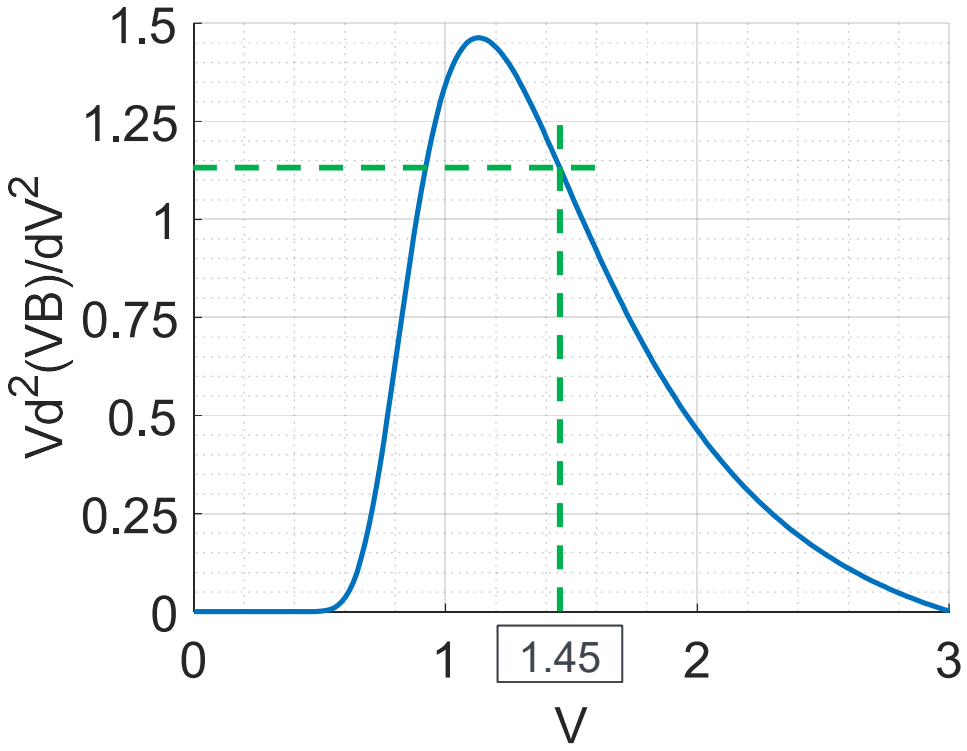
a)



Problem 10

University of Stuttgart
Germany

b)



Problem 11

University of Stuttgart
Germany

For an optical communication system, a graded index fiber is compared with a step index fiber. The graded index fiber has a parabolic refractive index profile.

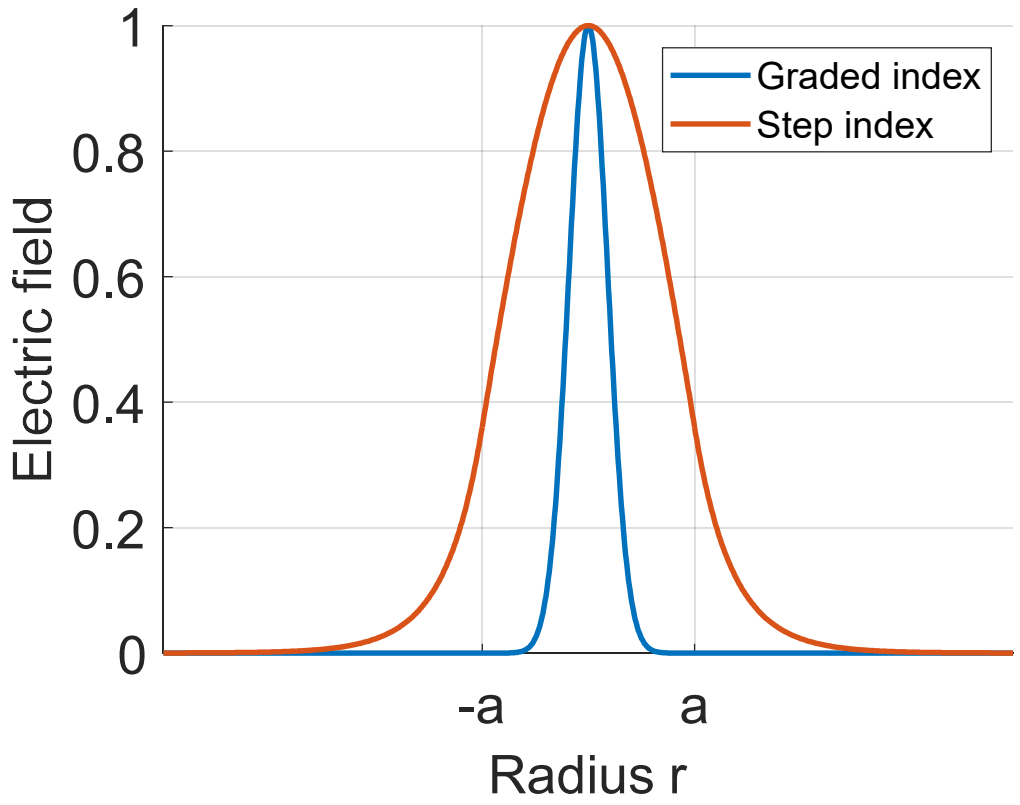
a) Find all supported modes for $V = 8$ for the step index fiber.

b) Assign the supported modes for the step index fiber to the corresponding mode groups of the graded index fiber. Are all modes of the step index fiber supported by the graded index fiber?

From now on, for both fibers is $V = 11$. The refractive indices are $n_1 = 1.45$ and $n_2 = 1.44$.

c) The used fiber has a length of 5 km. Calculate the differential mode delay for the 6th mode for the graded index fiber and the step index fiber. What are the maximum possible symbol rates?

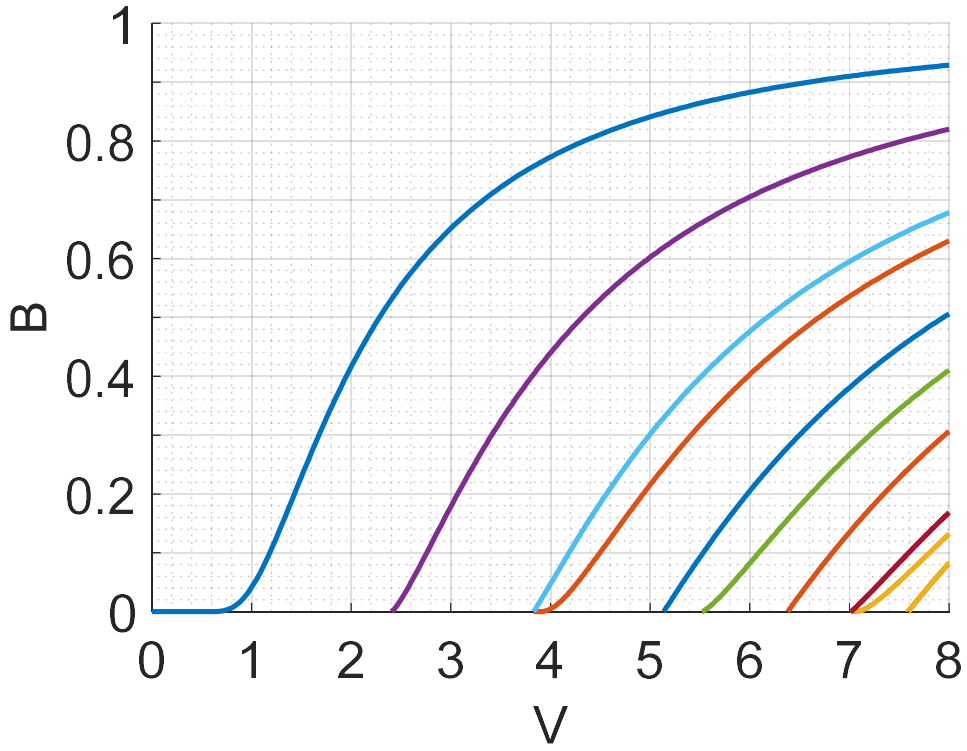
Problem 11


1
0.8
0.6
0.4
0.2
0
Electric field
-a
a
Radius r
Graded index
Step index

Problem 11

University of Stuttgart
Germany

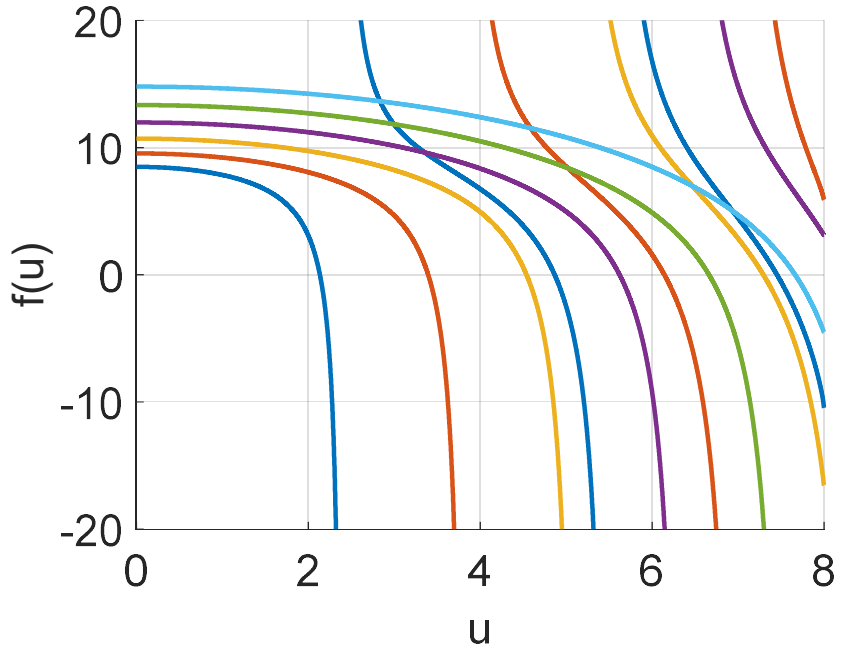
a)



Problem 11

University of Stuttgart
Germany

a)

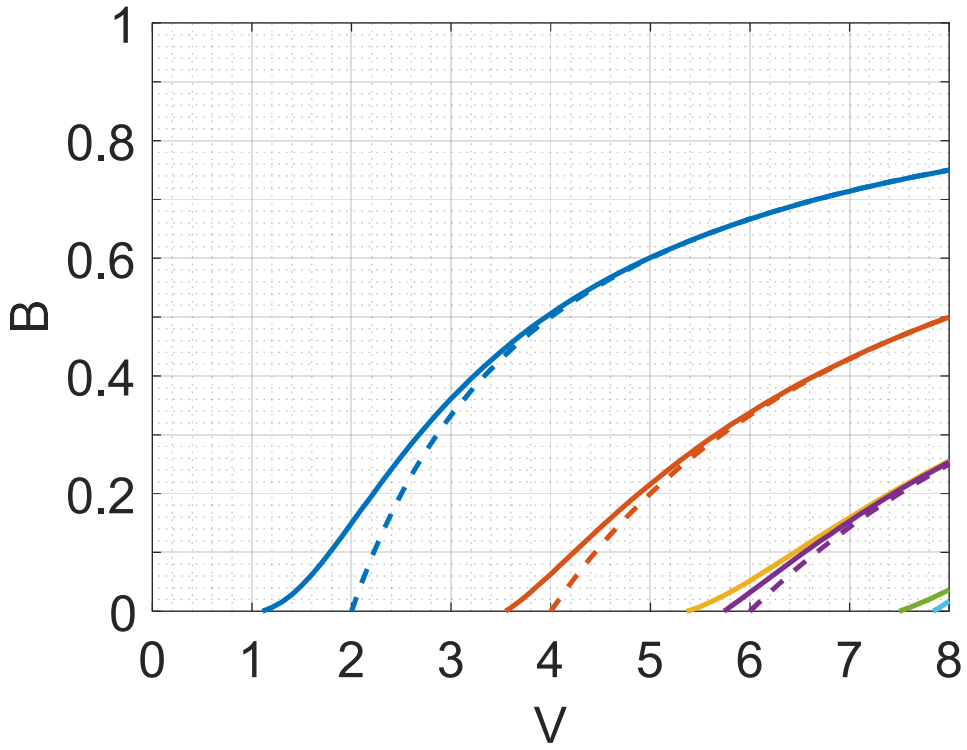


$$f(u)=\frac{uJ_{l-1}(u)}{J_l(u)}+\frac{vK_{l-1}\left(\sqrt{V^2-u^2}\right)}{K_l\left(\sqrt{V^2-u^2}\right)}$$

Problem 11

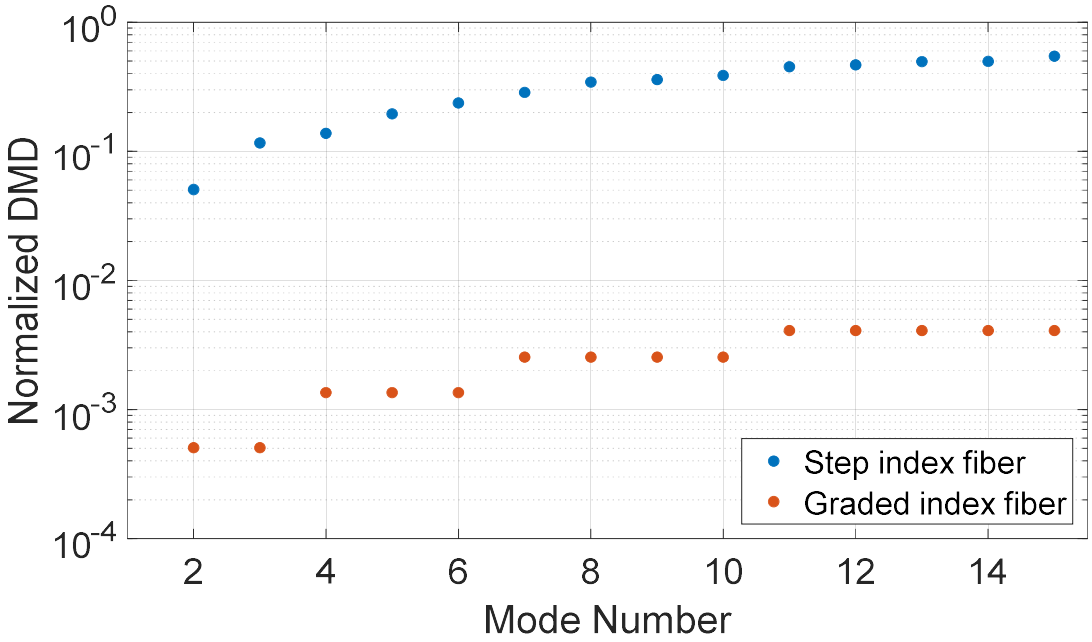
University of Stuttgart
Germany

b)



Problem 11

University of Stuttgart
Germany



Problem 12

University of Stuttgart
Germany

The light guided in a monomodal step index fiber shall be coupled into a rectangular waveguide on a photonic chip. The plane of coupling is directly at the edge of the chip. In order to achieve an efficient coupling, a fiber transition structure shall be investigated.

a) What is the coupling efficiency, when the light is coupled directly from the fiber to the rectangular waveguide? The radius of the fiber core is $a_1 = 3.2$ µm and the waveguide width is 0.4 µm.

b) In order to increase the coupling efficiency, the radius of the fiber core shall be decreased. Find the coupling efficiency for one, two, and four transitions.

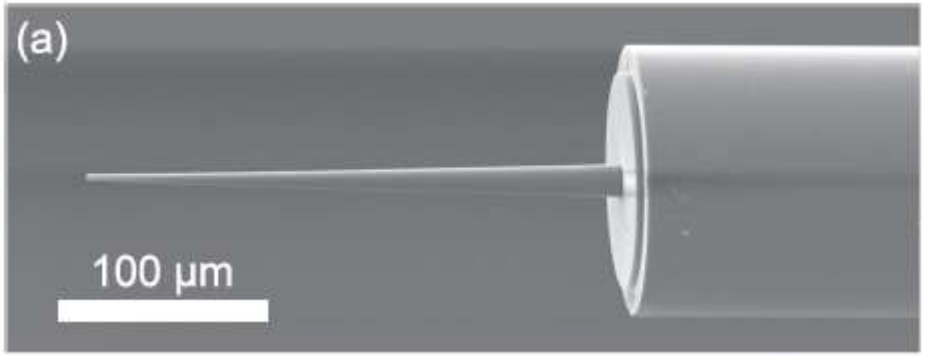
Problem 12

c) Express the radius of the terminal fiber core as a function of the incident fiber core radius a_1 and the constant ratio α for n transition steps and deduce α as a function of n. What is the form of the used transition?

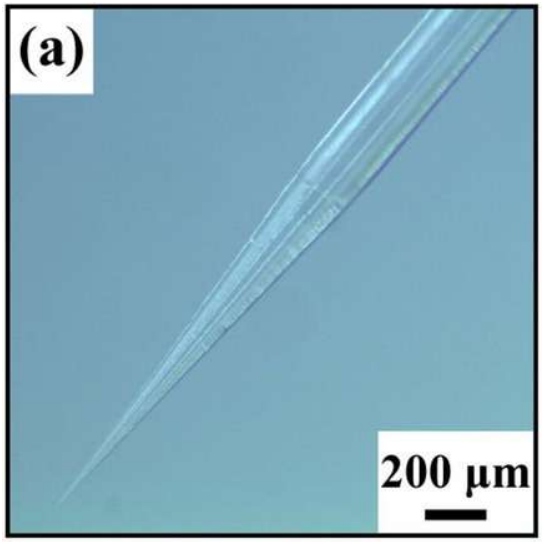
d) Express the coupling efficiency for n transition steps and calculate the coupling efficiency for $n = 60$.

e) How large is the achieved coupling efficiency for $n \rightarrow \infty$?

Problem 12



K. Vanmol et al., 2020



J. Cao et al., 2016

Thank you!



M.Sc. Julian Schneck

e-mail Julian Schneck@int.uni-stuttgart.de
phone +49 (0) 711 685 67919
www.int.uni-stuttgart.de/en

University of Stuttgart
Institute of electrical and optical communications



Survey

University of Stuttgart
Germany



https://evasysw.uni-stuttgart.de/evasys/online.php?p=TVKJJ